Assignment 1

Goal:

In this assignment you will turn a raw, messy laser powder bed fusion dataset into a clean, analysis-ready one. You will practice the three steps we covered in class, in order: cleaning the data (Week 3), exploring it with statistics and plots (Week 4), and building new features (Weeks 5 and 6).

Instructions for Students:

- You may use Python or Orange Canvas for this assignment.
- If using Python, you must set the random_seed based on your W-number for all data splits and model initializations.
- For example, if W-number is w1723axa use random_seed = 1723.
- You may use Generative AI to assist with writing code or navigating the Orange interface. However, your report must be your own analysis. To receive credit, all answers must directly reference the specific numerical outputs, plots, and implications generated by your unique seed.
- Your report must follow the exact number of questions provided in this assignment. Failure to follow the specified numbering will result in no points being awarded for the corresponding responses.
- Include your Full Name and W-Number in the header of every page of your report.
- Submit your report as a PDF file.
- If you use Python or MATLAB, submit all relevant code files along with your report. Comment your code clearly: briefly explain what each step does and why. Clear comments make it easier to give you credit for your work.
- If you use Orange, submit a separate file containing screenshots of your workflow and results. Do not include the screenshots file as part of the report.
- Clearly document any assumptions, preprocessing decisions, or other decisions you made during the analysis in your report.

- You may be randomly selected to explain your code or workflow at any point. Make sure you understand every step you submit. Students are allowed to use gen AI but they have to understand and explain their work

Assignment 1: Melt Pool Data (Laser Powder Bed Fusion)
Assignment 1.1: Data Cleaning: spot the junk rows and delete them, fix the Defect Type typos, remove the duplicate rows, fill the truly blank numbers with the median, and decide what to do with the blank Defect Type cells.
Assignment 1.2: Data Analysis: range, maximum, minimum, average and standard deviation of each variable; histograms and data distributions; a scatter plot of Laser Power against Melt Pool Width; and a box plot of Cooling Rate by Defect Type.
Assignment 1.3: Feature Engineering: build Energy Density = Laser Power ÷ (Scan Speed × Beam Diameter), bin Shielding Gas into Low and High, create a "Has Defect" Yes/No column, and drop the columns you no longer need.

Submission:

For 1.1: the cleaned dataset, plus a DOC or PDF with screenshots of your process (if you used Orange) or your code (if you used Python or MATLAB).
For 1.2: a DOC or PDF with your results and plots, plus screenshots of your process (Orange) or your code (Python or MATLAB).
For 1.3: the dataset with feature engineering done, a complete report covering all 3 sections, and another DOC or PDF with screenshots of your process (Orange) or your code (Python or MATLAB).

Rubric

Assignment 1	Tasks	Grade
Assignment 1.1		25
	Deleting Junk Rows	5
	Fix Defect type typos	3
	Remove Duplicate rows	3
	Fill the truly blank number with the median	4
	Fixing Blank Type Cells	7
	Submitted DOC/PDF/Code	3
Assignment 1.2		35
	Range of each variable	3
	Maximum	2
	Minimum	2
	Average	3
	Standard Deviation	3
	Histogram and Data distribution	9
	scatter plot of Laser Power against Melt Pool Width	5
	box plot of Cooling Rate by Defect Type	5
	Submitted DOC/PDF/Code	3
Assignment 1.3		40
	Build Energy Density = Laser Power ÷ (Scan Speed × Beam Diameter)	9
	Bin Shielding Gas into Low and High	6
	Has Defect" Yes/No	5
	Drop Column	6
	Final report	10
	Submitted DOC/PDF/Code	4